

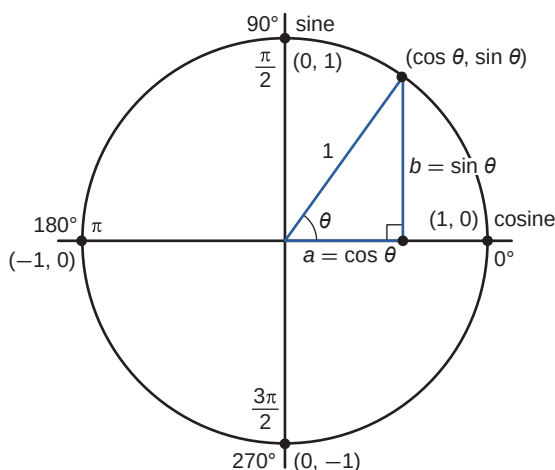


# Trigonometric Identities

Identities are useful for determining unknown trigonometric function values.

## Using the Pythagorean Identity

The triangle shown on the unit circle is a right triangle with leg lengths  $a = \cos \theta$  and  $b = \sin \theta$  and a hypotenuse of length 1.



By the Pythagorean theorem,  $a^2 + b^2 = 1$ . You can substitute  $\sin \theta$  for  $b$  and  $\cos \theta$  for  $a$  to get  $\sin^2 \theta + \cos^2 \theta = 1$ .

► **Think About It** The expression  $\sin^2 \theta$  is the same as  $(\sin \theta)^2$ , and  $\cos^2 \theta$  is the same as  $(\cos \theta)^2$ . The first way of writing the expressions is often used to emphasize that the entire function, not just the angle or input value, is being squared.

► **Think About It** The unit circle has a radius of 1 unit. The equation of the unit circle is therefore  $y^2 + x^2 = 1$ . This equation is another way that you can derive the equation that leads to the Pythagorean identity  $\sin^2 \theta + \cos^2 \theta = 1$ .

## Pythagorean Identity

For any angle  $\theta$ ,

$$\sin^2 \theta + \cos^2 \theta = 1.$$

This identity is called a **Pythagorean identity** because the equation that leads to it comes from the Pythagorean theorem.

### EXAMPLE 1

The angle  $\theta$  lies in Quadrant IV and  $\cos \theta = \frac{3}{4}$ . Determine the function value.

**A**  $\sin \theta$

#### SOLUTION

Use the Pythagorean identity.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta + \left(\frac{3}{4}\right)^2 = 1 \quad \text{Substitute } \frac{3}{4} \text{ for } \cos \theta.$$

$$\sin^2 \theta + \frac{9}{16} = 1 \quad \text{Square } \frac{3}{4}.$$

$$\sin^2 \theta = \frac{7}{16} \quad \text{Subtract } \frac{9}{16} \text{ from both sides.}$$

$$\sin \theta = \pm \frac{\sqrt{7}}{4} \quad \text{Square Root Property}$$

In Quadrant IV, sine is negative, so  $\sin \theta = -\frac{\sqrt{7}}{4}$ .

**B**  $\tan \theta$ **SOLUTION**

Use the equation  $\tan \theta = \frac{\sin \theta}{\cos \theta}$ .

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{-\frac{\sqrt{7}}{4}}{\frac{3}{4}} = -\frac{\sqrt{7}}{3} \quad \blacksquare$$

## Using Other Trigonometric Identities

### Properties

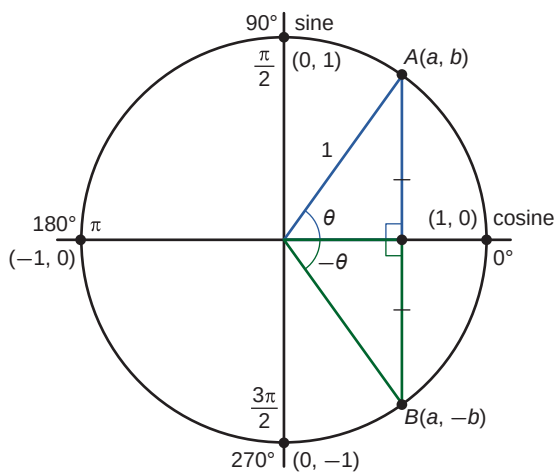
For any angle  $\theta$ ,

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

To see why the first two properties make sense, look at the unit circle. The angle  $-\theta$  has a measure that is the absolute value of the measure of angle  $\theta$ . Point  $A$  on the terminal side of  $\theta$  is  $(\cos \theta, \sin \theta)$ . Point  $B$  on the terminal side of  $-\theta$  is  $(\cos(-\theta), \sin(-\theta))$ .



The value of  $\cos(-\theta)$  is the same as the value of  $\cos \theta$  because both angles have the same horizontal coordinate. So  $\cos(-\theta) = \cos \theta$ .

However, the value of  $\sin(-\theta)$  is the opposite of the value of  $\sin \theta$  because the vertical coordinate of  $-\theta$  is the opposite of the vertical coordinate of  $\theta$ . So  $\sin(-\theta) = -\sin \theta$ .

## EXAMPLE 2

Determine  $\cos \theta$  when  $\sin(-\theta) = \frac{2}{5}$  and  $\tan \theta = \frac{2\sqrt{21}}{21}$ .

### SOLUTION

**Step 1** Determine  $\sin \theta$  using the property  $\sin(-\theta) = -\sin \theta$ .

$$\sin \theta = -\sin(-\theta) = -\frac{2}{5}$$

**Step 2** Solve  $\tan \theta = \frac{\sin \theta}{\cos \theta}$  for  $\cos \theta$ .

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cos \theta \cdot \tan \theta = \frac{\sin \theta}{\cos \theta} \cdot \cos \theta \quad \text{Multiply both sides by } \cos \theta.$$

$$\cos \theta \cdot \tan \theta = \sin \theta \quad \text{Simplify.}$$

$$\frac{\cos \theta \cdot \tan \theta}{\tan \theta} = \frac{\sin \theta}{\tan \theta} \quad \text{Divide both sides by } \tan \theta.$$

$$\cos \theta = \frac{\sin \theta}{\tan \theta} \quad \text{Simplify.}$$

**Step 3** Use  $\sin \theta = -\frac{2}{5}$  and  $\tan \theta = \frac{2\sqrt{21}}{21}$  to determine  $\cos \theta$ .

$$\cos \theta = \frac{\sin \theta}{\tan \theta} = \frac{-\frac{2}{5}}{\frac{2\sqrt{21}}{21}} = -\frac{\sqrt{21}}{5} \quad \blacksquare$$

# Proving Identities

To prove that an identity is true, transform one or both sides of the equation so that they are identical.

## Strategies for Proving Trigonometric Identities

Here are some tips to keep in mind when proving identities:

- Try to rewrite the more complicated side of the equation so that it is identical to the less complicated side. If both sides are fairly complicated, rewrite both sides so they are identical to the same simpler expression.
- Try writing  $\tan \theta$  in terms of  $\sin \theta$  and  $\cos \theta$ .
- If there is a mixture of  $\sin^2$  and  $\cos^2$ , then use the Pythagorean identity to get just one or the other.

### EXAMPLE 3

$$\sin^2 \theta \left( 1 + \frac{1}{\tan^2 \theta} \right) = 1$$

#### SOLUTION

Rewrite the left side so that it is identical to the right side.

$$\sin^2 \theta \left( 1 + \frac{1}{\tan^2 \theta} \right) = 1$$

$$\sin^2 \theta \left( 1 + \frac{1}{\frac{\sin^2 \theta}{\cos^2 \theta}} \right) = 1 \quad \text{Substitute } \frac{\sin \theta}{\cos \theta} \text{ for } \tan \theta.$$

$$\sin^2 \theta \left( 1 + \frac{\cos^2 \theta}{\sin^2 \theta} \right) = 1 \quad \text{Simplify.}$$

$$\sin^2 \theta + \cos^2 \theta = 1 \quad \text{Distribute.}$$

$$1 = 1 \quad \text{Pythagorean Identity} \quad \blacksquare$$